



Microsoft Defender for Endpoint

Master Class

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May 2026

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Agenda

- Module 1: Introduction Microsoft Defender for Endpoint
- Module 2: Onboarding Endpoints
- Module 3: Protecting Endpoints
- Module 4: Device Actions
- Module 5: Additional Configurations
- Module 6: KQL Introduction

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Module 01

Overview

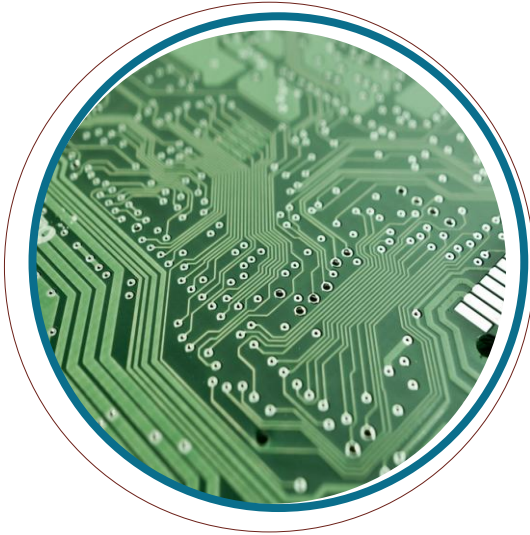
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Module 1 Contents:

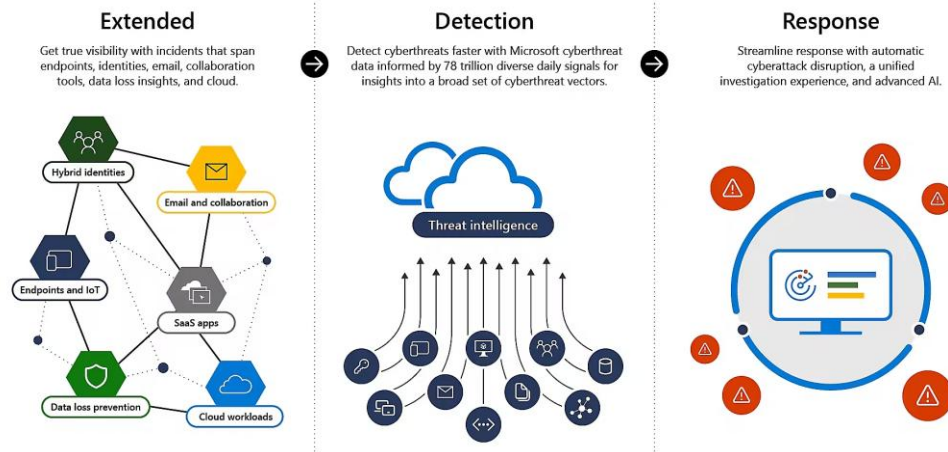
- **What is Microsoft Defender XDR?**
- **What is Microsoft Defender for Endpoint (MDE)?**
- **MDE vs. MS Intune**
- **Introduction to MDE**
 - Feature Overview
 - Licensing
 - Activating MDE
 - XDR/MDE portal
- **Role based access control**

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Microsoft XDR

Microsoft Defender XDR



Microsoft Defender XDR

- also: *Microsoft XDR*
- Platform for securing
 - Endpoints
 - Identities
 - Email and Collaboration
 - Cloud Apps
 - Compute resources
 - online
 - on-prem

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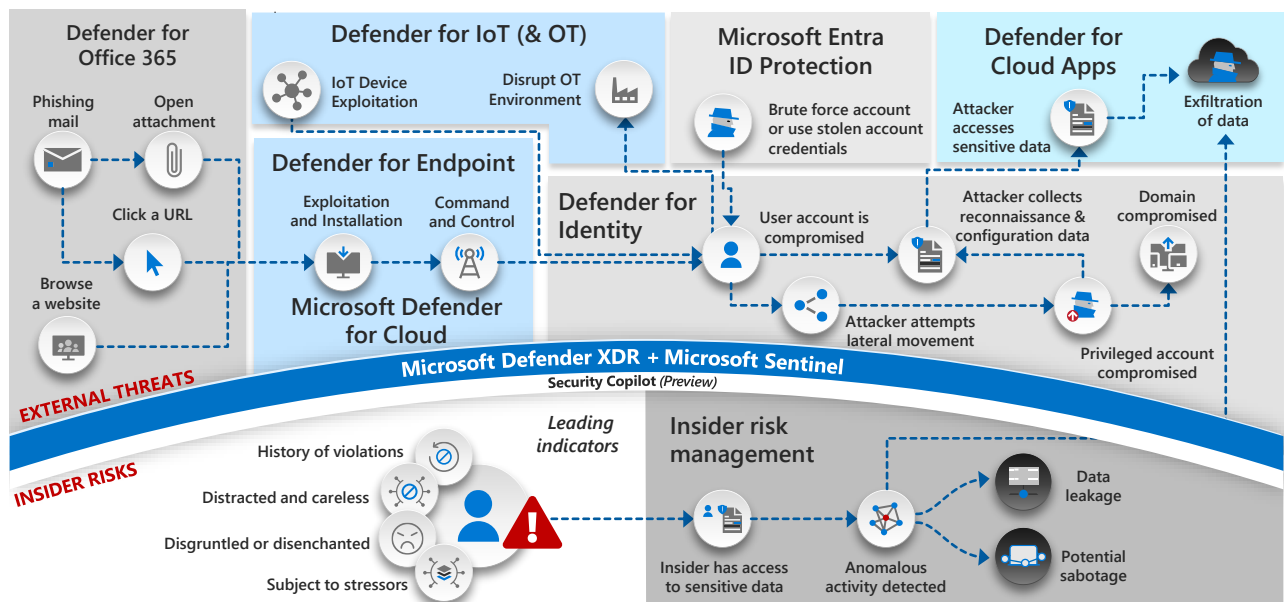
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Defend across attack chains

Insider and external threats

Microsoft December 2023 – <https://aka.ms/MCRA>



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Microsoft Defender for Endpoint

- *'... is an enterprise security platform designed to help enterprise networks prevent, detect, investigate and respond to advanced threats.'*^[1]
- Endpoint:
 - Laptops, phones, tablets, PCs
 - Access points, routers, firewalls

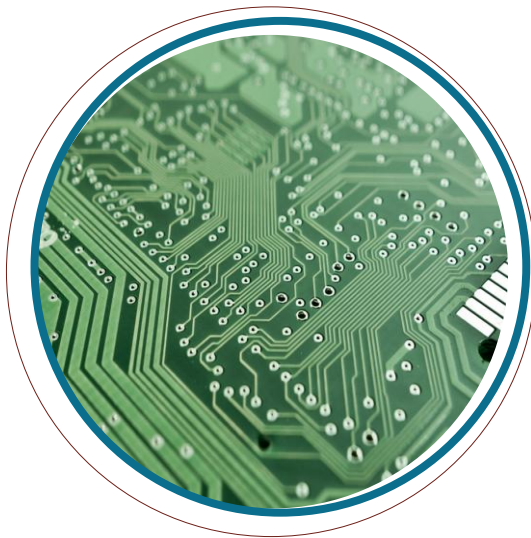
source: ^[1] <https://learn.microsoft.com/en-us/defender-endpoint/microsoft-defender-endpoint>

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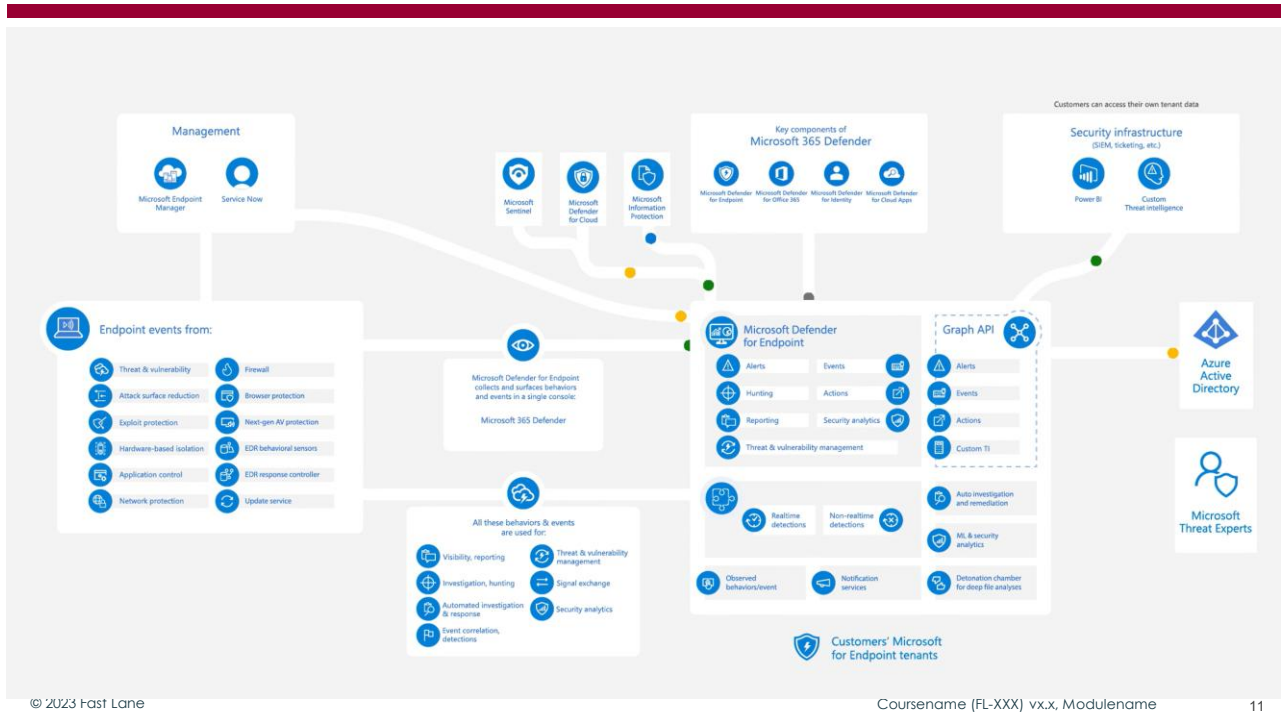
MDE Architecture

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MDE - Architecture

- Cloud/Portal



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MDE - Architecture

- Endpoint Sensors



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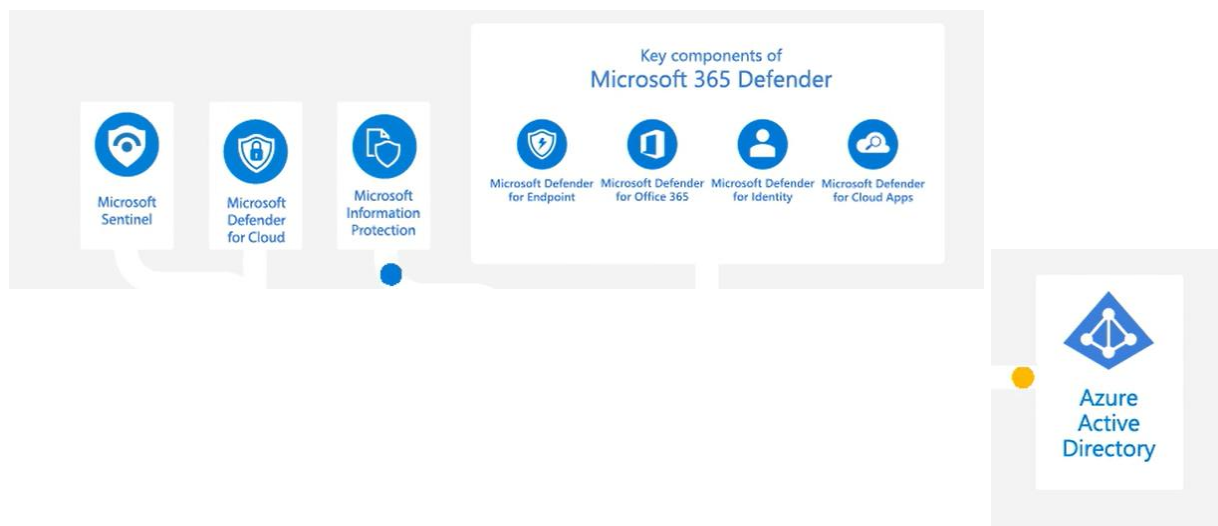
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MDE - Architecture



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MDE - Architecture



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MDE Portal

- Settings
- Device Inventory
- Threat & Vulnerability Management
- Incidents & Alerts
- Hunting
- Actions & submissions

The screenshot shows the Microsoft Defender MDE Portal interface. The left sidebar contains a navigation menu with the following items:

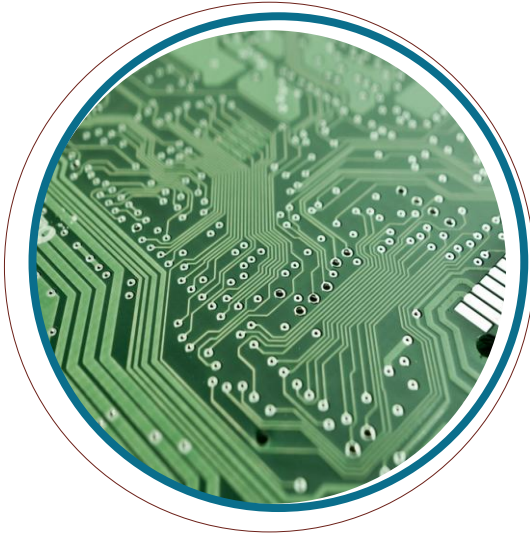
- Home
- Exposure management
- Investigation & response
- Incidents & alerts
- Hunting
- Actions & submissions
- Partner catalog
- Assets
- Devices
- Microsoft Sentinel
- Endpoints
- Vulnerability management
- Partners and APIs
- Configuration management
- Email & collaboration
- SOC optimization
- Reports
- System
- Audit
- Permissions
- Health
- Settings
- Customize navigation
- Show all

The main content area displays the "Home" page with the following sections:

- Get your SIEM and XDR in one place:** A banner encouraging users to connect Microsoft Sentinel and Microsoft Defender XDR for unified security search and threat intelligence, with a "Connect a workspace" button.
- SOC optimization:** A section showing optimization status with a bar chart for Active, In progress, Completed, and Dismissed states.
- Your optimizations data:** A section showing recent optimizations value, stating "No recent optimizations value" and explaining that values will be shown once optimizations are completed.
- Top threat-based coverage optimizations:** A section showing "No recent coverage optimizations" with a "Configure" button.

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MDE

Features

Licenses

Source: <https://www.microsoft.com/en-us/security/business/endpoint-security/microsoft-defender-endpoint>

MDE Features

MDE Plan 1

Unified security tools and centralized management
 Next-generation antimalware
 Cyberattack surface reduction rules
 Device control (such as USB)
 Endpoint firewall
 Network protection
 Web control/category-based URL blocking
 Device-based conditional access
 Controlled folder access
 APIs, SIEM connector, custom threat intelligence
 Application control

MDE Plan 2

Endpoint detection and response
 Deception techniques
 Automated investigation and remediation
 Cyberthreat and vulnerability management
 Threat intelligence (cyberthreat analytics)
 Sandbox (deep analysis)
 Endpoint attack notifications

Source: <https://learn.microsoft.com/en-us/office365/servicedescriptions/microsoft-365-service-descriptions/microsoft-365-tenantlevel-services-licensing-guidance/microsoft-365-security-compliance-licensing-guidance#microsoft-defender-for-endpoint>

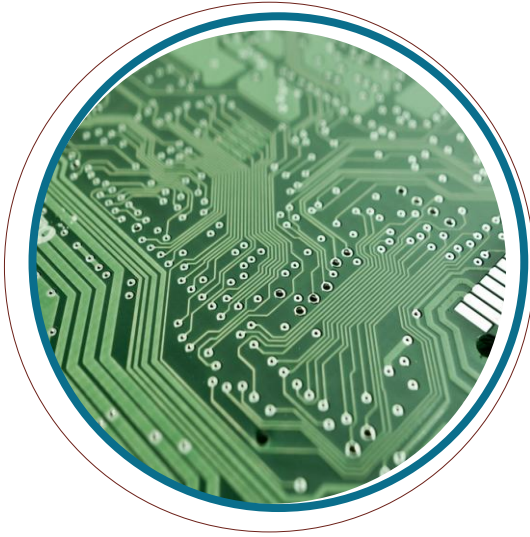
MDE Licenses

- **MDE Plan 1**
 - standalone user subscription license
 - part of Microsoft 365 E3/A3/G3
- **MDE Plan 2**
 - standalone user subscription license
 - part
 - Microsoft 365 E5/A5/G5 (Windows 11 Enterprise E5 incl.)
 - Microsoft E5/A5/G5/F5 Security
 - Microsoft 365 F5 Security & Compliance
 - Windows 11 Enterprise E5/A5

Source: <https://learn.microsoft.com/en-us/office365/servicedescriptions/microsoft-365-service-descriptions/microsoft-365-tenantlevel-services-licensing-guidance/microsoft-365-security-compliance-licensing-guidance#microsoft-defender-for-business>

MDE Licenses

- Microsoft Defender for Business
 - standalone user license
 - part of Microsoft Business Premium
 - features of MDE plan 1 plus some of plan 2
 - server add-on available
- Microsoft Defender for Servers
 - part of Microsoft Defender for Cloud
 - Azure subscription required
 - also suitable for hybrid or multi-cloud hosted servers



MDE / Microsoft Intune

MDE vs. Microsoft Intune

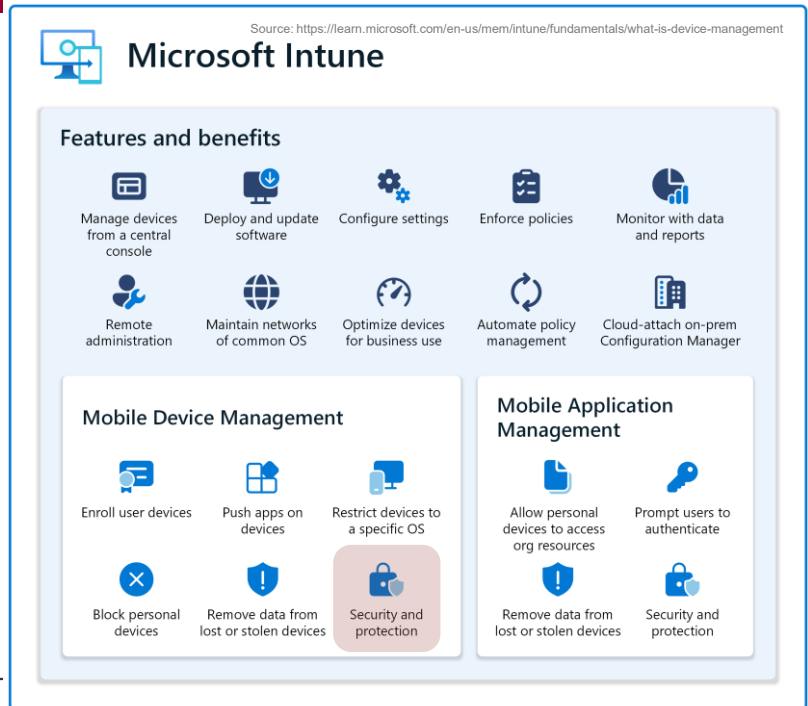
'Microsoft Intune is a **cloud-based endpoint management solution**. It manages user access to organizational resources and simplifies app and device management across your many devices, including mobile devices, desktop computers, and virtual endpoints.'

MDE vs. Intune

'Microsoft Intune is a **cloud-based endpoint management solution**. It manages user access to organizational resources and simplifies app and device management across your many devices, including mobile devices, desktop computers, and virtual endpoints.'

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MDE vs. Microsoft Intune

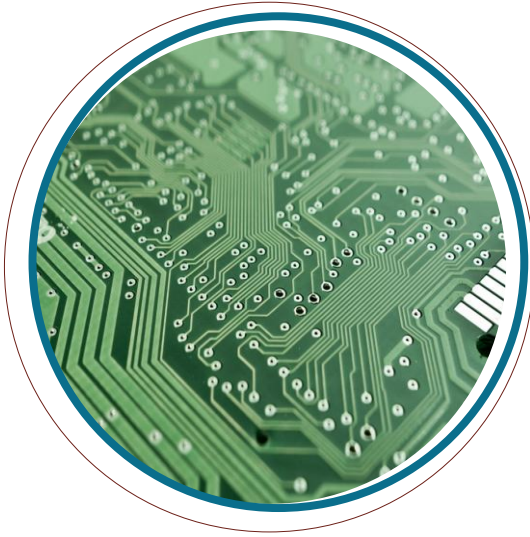
- Intune is not required for MDE
- MDE is not required for Intune
- MDE extends security capabilities on and for devices
 - EDR
 - Security configuration
- Intune portal could be used for some settings in MDE

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Zero Trust

Source: <https://learn.microsoft.com/en-us/defender-endpoint/zero-trust-with-microsoft-defender-endpoint>

Zero Trust

'... is a security strategy for designing and implementing the following set of security principles:'

Verify explicitly	Use least privilege access	Assume breach
Always authenticate and authorize based on all available data points.	Limit user access with Just-In-Time and Just-Enough-Access (JIT/JEA), risk-based adaptive policies, and data protection.	Minimize blast radius and segment access. Verify end-to-end encryption and use analytics to get visibility, drive threat detection, and improve defenses.

Defender for Endpoint is a primary component of the Assume breach principle and an important element of your extended detection and response (XDR) deployment with MS Defender XDR.

Zero Trust

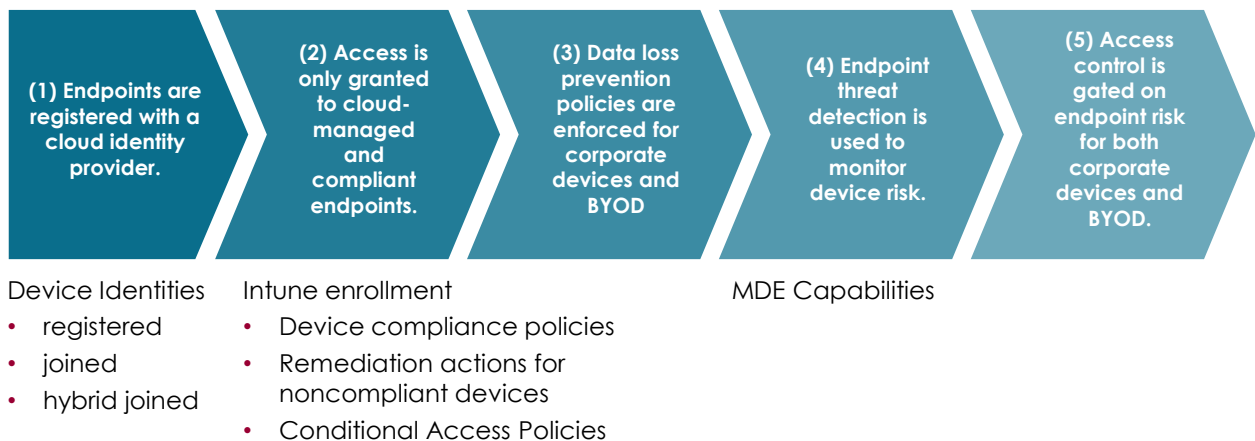
used technologies

- Endpoint behavioral sensors
- Cloud security analytics
- Threat intelligence

Threat protection for Zero Trust

- Core Defender Vulnerability Management
- Attack surface reduction
- Next-generation protection
- EDR
- Automated investigation and remediation
- Secure Score for Devices
- MS Threat Experts

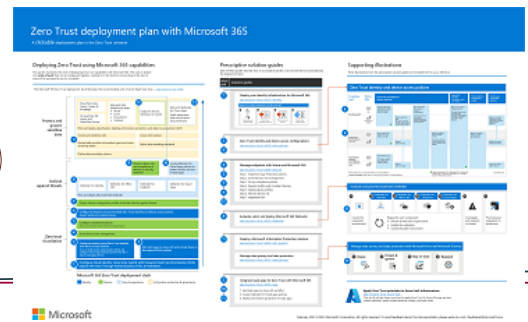
Endpoint Zero Trust deployment guide





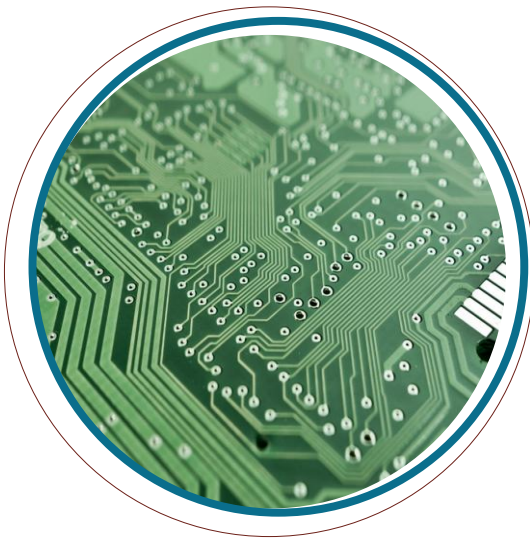
MDE Zero Trust Deployment guide is part of

Zero Trust Deployment Guide



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MDE Deployment

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Source: <https://learn.microsoft.com/en-us/defender-endpoint/production-deployment>

MDE Deployment (Step 1)

Set up

- Check licenses
- use Microsoft Defender portal
 - <https://security.microsoft.com>
 - click on Assets/Devices (e.g.)
 - for the first time we have to go for a coffee
 - wait for Org ID in Settings | Microsoft Defender XDR
 - and the item Settings | Endpoints
- Data center location
 - same as for Defender XDR
 - cannot be changed



Hang on! We're preparing new spaces for your data and connecting them.



This takes a few minutes. When we're done, your data will gradually consolidate and light up the console in the next few hours. [Learn about Microsoft Defender XDR](#)

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Source: <https://learn.microsoft.com/en-us/defender-endpoint/production-deployment>

MDE Deployment (Step 2)

Assign roles and permissions

- Concept of least Privileges
- Choices
 - Basic permissions management
 - RBAC
 - Unified RBAC (P2 required)
- more later
- for the start be a Global or Security Administrator (Entra ID role)

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MDE Deployment (Step 3)

Identify architecture and deployment/onboarding method

- Know the OS(s) of devices
- Know the current configuration management system
 - Config Manager
 - Intune
 - GPO
 - ...
- Derive the onboarding method

MDE Deployment (Step 3)

Identify architecture and deployment/onboarding method

Architecture	Description
Cloud-native	We recommend using Microsoft Intune to onboard, configure, and remediate endpoints from the cloud for enterprises who don't have an on-premises configuration management solution or are looking to reduce their on-premises infrastructure.
Co-management	For organizations who host both on-premises and cloud-based workloads we recommend using Microsoft's ConfigMgr and Intune for their management needs. These tools provide a comprehensive suite of cloud-powered management features, and unique co-management options to provision, deploy, manage, and secure endpoints and applications across an organization.
On-premises	For enterprises who want to take advantage of the cloud-based capabilities of Microsoft Defender for Endpoint while also maximizing their investments in Configuration Manager or Active Directory Domain Services , we recommend this architecture.
Evaluation and local onboarding	We recommend this architecture for SOCs (Security Operations Centers) who are looking to evaluate or run a Microsoft Defender for Endpoint pilot, but don't have existing management or deployment tools. This architecture can also be used to onboard devices in small environments without management infrastructure, such as a DMZ (Demilitarized Zone).

MDE Deployment (Step 3)

Identify architecture and deployment/onboarding method

- Deployment tool
 - local script
 - GPO
 - MS Intune
 - MS Config Manager
 - VDI scripts
 - ...

MDE Deployment (Step 3)

Identify architecture and deployment/onboarding method

Endpoint	Deployment tool	
Windows	Local script (up to 10 devices) Group Policy Microsoft Intune/ Mobile Device Manager	Microsoft Configuration Manager VDI scripts
Win/Lx servers	Integration with Microsoft Defender for Cloud	
macOS	Local script Microsoft Intune JAMF Pro	Mobile Device Management
Linux servers	Local script Puppet Ansible	Chef Saltstack
Android	Microsoft Intune	
iOS	Microsoft Intune Mobile Application Manager	

MDE Deployment (Step 4)

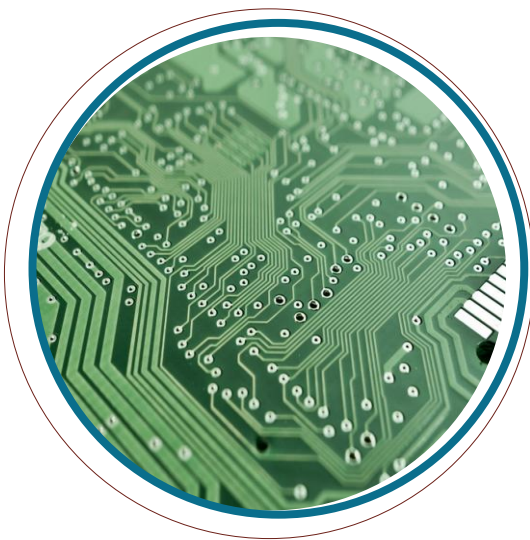
Onboard Devices

- discussed later ...

MDE Deployment (Step 5)

Configure MDE capabilities

- discussed later ...



Permissions

MDE Permissions

Basic Permissions

- Default Entra ID Roles

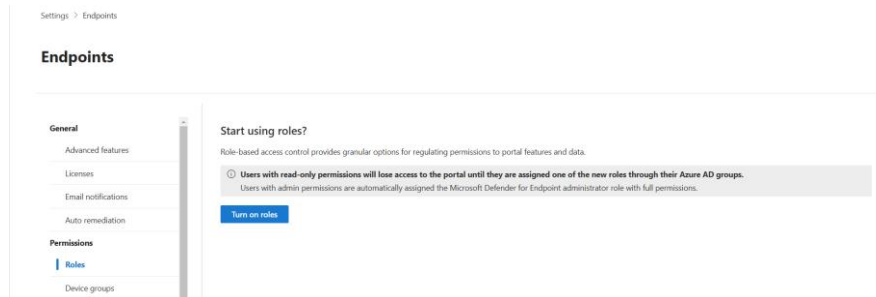
Global Administrator Security Administrator	Full Access
Security Reader	Read-only access No Access to device inventory

MDE Permissions

Role-based access control (RBAC)

- Configured in MDE settings
- Roles & Device groups
- suitable if tiering is necessary
- Turn on
Settings | Endpoints | Permissions/Roles
Settings | Endpoints | Permissions/Device groups

MDE Permissions



- Turn on
 - XDR Portal | Settings | Endpoints | Permissions/Roles
 - irreversible
 - former admins became MDE administrators with full permissions
 - former readers will have no permissions

MDE Permissions - RBAC

- Roles are
 - a set of permissions
 - assigned to an Entra ID group
 - stored under a custom name
- Device groups are
 - a set of onboarded devices
 - devices are filtered by name, domain, tag, OS (and/or)
 - assigned to an Entra ID group with assigned role

MDE Permission - Unified RBAC

'The Microsoft Defender XDR Unified role-based access control (RBAC) model provides a single permissions management experience that provides one central location for administrators to control user permissions across different security solutions.'

- Cross workload permissions within a single role
- Ability to assign roles to individual users as well as security group
- Multi assignment within a single role

MDE Permission - Unified RBAC

Process

- Assure to be a Global or Security Administrator
- Using
 - Create custom roles
 - Import existing roles
 - View, edit and delete
- Activate MS XDR Unified RBAC model

Microsoft Defender

Try the Microsoft Defender XDR permission model and get more granular control.

Permissions

Admins are users added to the admin role, and can view any data and complete any task in the Microsoft Defender portal. You should have few admins, and grant users only the permissions needed to perform their duties when you assign access.

6 items

Name	Description
Microsoft Defender XDR (1)	
Roles	Manage who has access to view tasks and take actions in Microsoft Defender XDR. Assign only the minimum permissions that admins need to get the job done.
Microsoft Entra ID (1)	
Roles	Manage who has access to view tasks and take actions in the Microsoft Defender portal. Assign them only the permissions needed to perform their duties.
Endpoints roles & groups (2)	
Roles	Manage endpoint access to view content and perform tasks. Grant only the permissions needed for users to perform their tasks.
Device groups	Organize devices into groups, set automated remediation levels, and assign administrators as needed.
Cloud Apps (2)	
Roles	Manage who has access to view cloud apps content and take actions. Assign them only the permissions needed to perform their duties.
Activity Privacy Permissions	Control which admins can reveal private activities.

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MDE Permissions - Unified RBAC

Role settings

- Name
- Description
- Permission groups
 - Security operations
 - Security posture
 - Authorization and settings

Choose permissions

Select permissions from each permission group to customize this role.

3 items

Permission group	Description
Security operations None selected	Manages day-to-day operations and responds to incidents and advisories
Security posture None selected	Manages the organization's security posture, performs Defender Vulnerability Management
Authorization and settings None selected	Manages the security and system settings, creates and assigns roles

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Permission Groups

Security operations

Select the permissions in this group to users who perform security operations and those who respond to incidents and advisories.

✕ Clear all permissions

- ☐ All read-only permissions
☐ All read and manage permissions
☒ Select custom permissions

Security data

- ☐ Read-only
☐ Select all permissions
☒ Select custom permissions
- ☐ Security data basics (read)
 - ☐ Alerts (manage)
 - ☐ Response (manage)
 - ☐ Basic live response (manage)
 - ☐ Advanced live response (manage)
 - ☐ File collection (manage)
 - ☐ Email & collaboration quarantine (manage)
 - ☐ Email & collaboration advanced actions (manage)

Raw data (Email & collaboration)

- ☐ Read-only
☒ Select custom permissions
- ☐ Email & collaboration metadata (read)
 - ☐ Email & collaboration content (read)

Security posture

Select the permissions for users who need to act on security recommendations, and track remediation tasks, exceptions, and vulnerabilities.

✕ Clear all permissions

- ☐ All read-only permissions
☐ All read and manage permissions
☒ Select custom permissions

Posture management

- ☐ Read-only
☐ Select all permissions
☒ Select custom permissions
- ☐ Vulnerability management (read)
 - ☐ Exception handling (manage)
 - ☐ Remediation handling (manage)
 - ☐ Exposure Management (read)
 - ☐ Exposure Management (manage)

Authorization and settings

Select the permissions for users who need to configure your security and system settings, and create and assign roles.

If you select any permissions on this page, you will also assign the security data read permission under the Security operations permission group.

✕ Clear all permissions

- ☐ All read-only permissions
☐ All read and manage permissions
☒ Select custom permissions

Authorization

- ☐ Read-only
☒ Read and manage

Security settings

- ☐ Read-only
☐ Select all permissions
☒ Select custom permissions
- ☐ Detection tuning (manage)
 - ☐ Core security settings (read)
 - ☐ Core security settings (manage)

System settings

- ☐ Read-only (Defender for Office, Defender for Identity)
☒ Read and manage

MDE Permission

References

- [Unified RBAC permissions](#)
- [Map Unified RBAC permissions to existing RBAC permissions](#)



End of Module 1